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An Open Dataset for the Acoustic Monitoring of Nocturnal Migratory Birds in Europe

Louis Airale[1,3] ✉, Adrien Pajot[1,3] ✉ & Juliette Linossier[2]

The persisting threats on migratory bird populations highlight the urgent need for effective monitoring techniques that could assist in their conservation. Among these, passive acoustic monitoring is an essential tool, particularly for nocturnal migratory species that are difficult to track otherwise. This work presents the Nocturnal Bird Migration (NBM) dataset, a collection of 13,359 annotated vocalizations from 117 species of the Western Palearctic, compiled through a crowd-sourcing effort. The dataset includes precise time and frequency annotations gathered by dozens of bird enthusiasts across France, enabling the automatic extraction of vocalizations in audio recordings and, further, novel downstream acoustic analysis. This comprises the development of a novel two-stage deep object detection model optimized for audio data, achieving competitive accuracy on the 45 most represented species, comparable to state-of-the-art systems trained on substantially larger datasets.

Background

Migratory birds have experienced a significant decline in numbers over recent decades due to various threats[1–5]. In light of the urgent need to conserve these threatened species, assessing their spatial and temporal distributions is crucial. Currently, several methods are used to monitor bird migration, including long-term monitoring at key sites, bird ringing, citizen science initiatives or biologging[6–12]. However, a significant fraction of bird species migrate at night[13–17]. Although biologging can track individual birds regardless of the time of day, methods relying on human observation and bird ringing are typically less effective or implemented on a reduced scale during nocturnal periods. At night, radars can estimate the number of individuals migrating each night in certain regions[18–21], but they cannot identify precisely the species in motion.

A promising solution for monitoring nocturnal migratory birds is the use of acoustic recorders that capture bird vocalizations, formally known as Passive Acoustic Monitoring (PAM)[22–24]. This method has gained increasing popularity and has become more widespread in recent years among passionate ornithologists, particularly during the COVID-19 pandemic, facilitated by the availability of recording devices for a general audience[25].

However, analyzing several hours of recordings to detect, identify, and count migrating birds is time-consuming and requires a high level of expertise. Automating the detection and classification by leveraging the recent and spectacular advances in computational image and sound processing is a promising avenue that has already provided compelling results. This first concerns wide-spectrum classification methods[26–28] trained on extensive but only weakly annotated datasets[29,30], where annotations do not indicate the precise signal occurrence within samples which often contain many background vocalizations. This introduces biases that can harm the classification accuracy of recognition models[31], or hinder their generalizability to potentially interesting use cases, such as automatic annotation of new data samples.

Another class of models trades data volume or species coverage for finely annotated data collections[32–41], where the scope is typically restricted to certain geographical locations, species or types of vocalizations. Part of these works also focuses on time annotations only, completely overlooking the precious information carried by frequency. Consequently, aside from rare exceptions such as the work of Shrestha *et al.*[42], bird sound recognition is consistently formulated as a *multi-label classification* task on sliding audio windows[34,43]. We argue however that addressing it as an *object detection* problem where vocalizations must be localized both in time and frequency may enable unique applications, such as monitoring various calling patterns or differentiating individuals from

[1]NBM-Nocturnal Bird Migration 404 chemin du casino Saint-Clément-des-Baleines, Saint-Clément-des-Baleines, 17590, France. [2]Biophonia NBM-Nocturnal Bird Migration Sualello Oletta, Sualello Oletta, 20232, France. [3]These authors contributed equally: Louis Airale, Adrien Pajot. ✉e-mail: laerien@gmail.com; pajot.adrien@wanadoo.fr

the same species. Indeed, while conspecific calls often overlap and sound similar, subtle individual differences in frequency contours, harmonics, or call timing may still be detectable in the time-frequency domain, allowing for the separation and enumeration of individual callers, even within the same species[44]. This is especially true of nocturnal migratory species, which tend to emit short, well-defined calls in a generally clear acoustic environment. Training an object detection model on bird calls however requires a large amount of finely annotated data which are crucially missing, especially in the case of European birds which are the primary focus of the present study. To overcome these challenges, the NBM project (for Nocturnal Bird Migration) was launched in 2020 to leverage the benefits of a crowd-sourcing initiative[45]. The objective was to share the annotation effort over privately collected recordings, complying with a standardized annotation procedure. Nighttime calls of migratory birds were targeted and annotated in time and frequency, allowing for the downstream training of fine-grained object detection models.

This article introduces the result of this work, the NBM-Dataset, a collection of 13,359 meticulously annotated vocalizations, primarily capturing nocturnal flight calls and songs from 117 bird species. To demonstrate the practical utility of this annotation approach, we develop a two-stage object detection model designed to precisely localize bird vocalizations on spectrograms. The model is based on a FPN Faster-RCNN architecture[46,47], with two key adaptations for sound event detection: we incorporate an attention mechanism to leverage the broader acoustic context of a signal, and add positional encodings to characterize the frequency and temporal scale of each sound event, as spatial invariance is not desirable in this context.

The resulting NBM detection model shows very competitive results with the BirdNet multi-label classification model[26], despite being trained on a much smaller database. This highlights the interest of our finely annotated dataset for diverse applications, ranging from general-purpose multi-label classification to the various tasks enabled by the precise localization of calls in time and frequency, such as the study of the variability of calling patterns. It is finally an additional brick in favor of future attempts to train data-consuming bioacoustic foundation models, which could provide the next breakthrough in the field. The NBM dataset is freely available under a permissive license[48]. The code and model are also accessible on the same page.

Related Work. General-purpose avian acoustic databases can be broadly segmented into two categories regarding the completeness of their annotated information.

Large data collections such as Xeno-Canto (xeno-canto.org, hereafter XC)[29] or the Macaulay library (macaulaylibrary.org)[30] contain respectively around 1 and 2 million weakly-annotated audio samples, for thousands of recording hours of bird species from all over the world. Although their labels provide no indication of the localization of the vocalizations they contain, making them unsuitable for the training of object detection systems, their large size is essential for the pre-training of many bird sound recognition models[26–28,49,50].

In parallel, a number of smaller-scale datasets featuring precise time and frequency annotations have been proposed in the wake of the annual BirdCLEF challenge[35–40,49,51], that typically contain tens of thousands of labels. As the cost of the annotation process is increased, the scope of these datasets is usually reduced in terms of species and geographical location. The present work aims to complete this picture, with a focus on nocturnal bird calls of the Western Palearctic. Finally, several other datasets contain localized annotations (see, for instance,[32,33,52–54], along with the great collection of databases of[55]) but these either omit the frequency information, lack species labels, species than the ones targeted in this study.

Methods

The NBM project was launched to build a comprehensive database of vocalizations from migratory birds active at night, designed to be easily accessible for both bird enthusiasts and computer scientists. The initiative also aimed to encourage contributions through a simple and unified annotation process. A community of bird experts soon came together, sharing recordings of nighttime bird sounds along with the corresponding annotations, which provided the necessary data to develop automated systems capable of replicating the experts' annotation methodology.

This essentially implies precisely marking bird sound *patterns* both in time and frequency on a spectrogram using the Audacity software[56]. These patterns were defined as the smallest identifiable segments required to recognize specific bird species, from either calls or songs. This method shares similarities with the 'phrases' concept used in BirdDB[52].

While there was naturally some room for personal judgment due to the diversity and complexity of bird vocalizations, as well as varying recording conditions, the specificity of nighttime soundscape—characterized by more calls than songs and fewer anthropogenic noises—helped reduce the annotation variance.

Leveraging the French Birding Community. The original goal was to compile at least 100 calls from the 60 most recorded migratory bird species in Europe. Recognizing the heavy workload involved in annotating thousands of vocalizations, we decided to involve the French birdwatching community in a collective annotation effort. A call to action was issued in December 2020 and the webinar introducing the project attracted 50 participants eager to assist with the annotations. Following that, a web platform (nocturnal-bird-migration.com/fr) was developed where participants could submit audio and annotations.

Since the project's launch, dozens of volunteers have contributed 11,381 annotated vocalizations from 97 species. While the majority of them are migratory birds, the collection also includes nocturnal singers, such as those in the Strigidae family, and other common species that can vocalize at night.

Data Curation. The web platform which received each new file upload was developed to streamline data management, unify and centralize the process of data collection. Notably, it featured an automatic validation system ensuring format and annotation quality, composed of the following five steps:

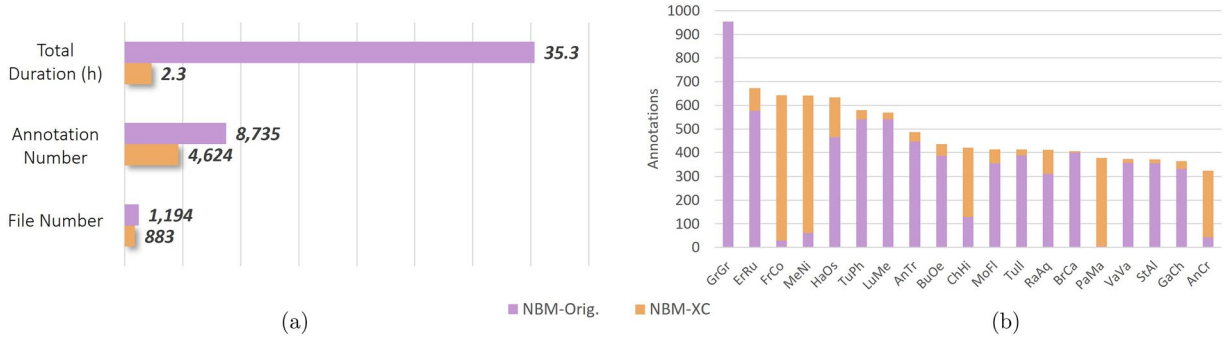


Fig. 1 **(a)** Data volume breakdown between the original database and the XC extra base. **(b)** Annotation repartition between the original and XC bases for the 20 most common species. We refer the reader to Table 2 for the correspondence between short and proper Latin names.

- Consistent matching of filenames between audio and annotation files.
- Proper file formats, with audio files in .wav and annotations in .txt.
- Adherence to Audacity's text file labeling standards.
- Presence and correct ordering of frequency and timestamp annotations.
- Use of standardized scientific names of the TAXREF v14.0 taxonomy[57], preventing typographical and taxonomic errors.

While these automated checks ensured uniformity and saved time, they did not account for the precise positioning of bounding boxes or species identification accuracy. Consequently, a thorough manual review was conducted to verify bounding box correctness and make necessary adjustments. A bounding box was considered correct if it tightly enclosed the full visible extent of a single vocalization event on the spectrogram, both in time and frequency, without including unrelated signals or excessive background noise. This criterion ensured consistent and precise localization of sound events for model training and evaluation.

Addressing the Dataset Biases. Involving a community of annotators to collect and annotate the data allowed to tackle a considerable dataset compilation task. While the resulting heterogeneity in recording conditions and equipments can enhance the robustness of downstream machine learning models, we also observed a number of biases arising from an imbalanced distribution of annotations across bird species and individual contributors. Common bird species were disproportionately represented, and some recording locations had an overabundance of annotations for just a few species, risking a skewed focus on a limited set of individuals in the database. To address this, we supplemented the dataset with additional audio samples from the Xeno-Canto database[29], emphasizing the need for greater intra-class variability and better balance between species. Since XC recordings typically include only a single class label but often contain vocalizations from multiple species, we conducted a manual time and frequency annotation process to align these samples with the standards of our database. The resulting distribution of sample origins for the 20 most common species is shown in Fig. 1 (right), illustrating our endeavor to achieve a more balanced and diverse database.

Data Records

Database Structure. The dataset is available on a Zenodo repository[48]. It is organized into two primary directories: *train* and *test*. Within the *train* directory, the data is further divided into two subdirectories: *train_nbm_orig* (files originally recorded by the participants) and *train_nbm_xc* (files downloaded from Xeno-Canto).

Each audio file, formatted in WAV, is accompanied by a name-matching text file (TXT) containing the labels. These labels adhere to Audacity formatting standards, and contain both time and frequency annotations, along with the scientific bird name. The naming convention for audio files sourced from Xeno-Canto includes the species name and the file ID from the database.

Database properties. The main figures regarding the properties of the NBM database are reported in Fig. 1 (left), together with the share represented by its two components, the original NBM base (NBM-Orig.) and the files originating from XC (NBM-XC), in terms of recording duration, file, and annotation number. The overall database contains almost 38 hours of recording, while total file and annotation numbers amount respectively to 2,077 and 13,359 for a total of 117 bird species. Among these, 86 species possess at least 10 annotated calls and 56 species at least 100. Additionally, 63 species have been annotated across at least 10 distinct files and 16 in more than 50 files.

NBM Test Set. Along with the training dataset described above, we introduce a test set to facilitate future research and monitor the improvements in both data collection and machine learning recognition models. The scope of the test set is restricted to the 45 species for which we considered the amount of training data to be sufficient (100 sound events over 20 distinct recordings, see Model Evaluation section for details). Six audio files were extracted for each species from XC and manually annotated, for a total of 270 test files. To prevent data leakage between the training and test sets, we only selected recordings that were uploaded after the collection of the NBM

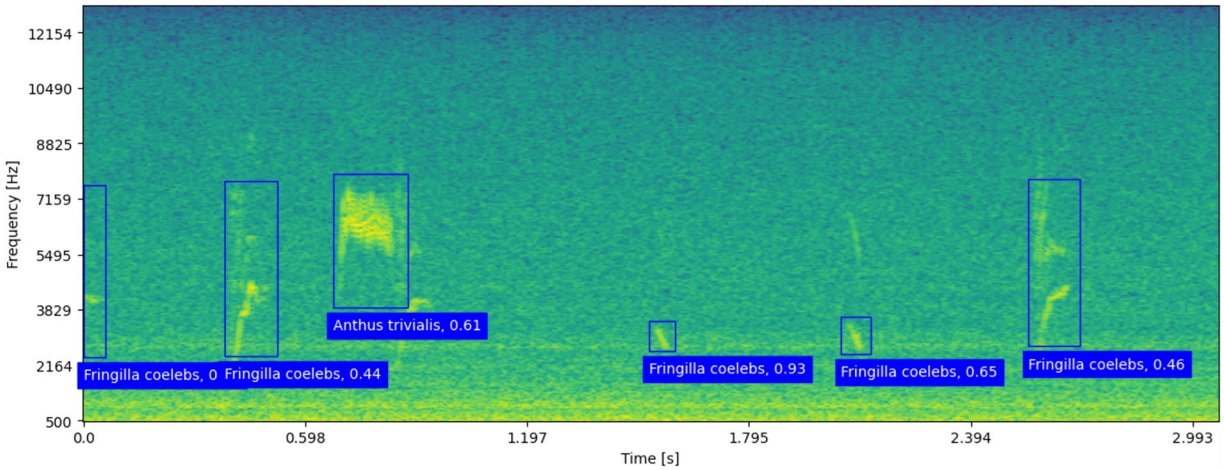


Fig. 2 The object detection approach allows a fine-grained time and frequency localization of short vocalizations, with applications in the study of calling patterns. In this example, the y-axis coordinates of the bounding boxes allow to distinguish the standard flight call of the Common chaffinch *Fringilla coelebs* from the ascending call emitted when the bird is perched (model confidence scores are reported along with predicted species).

training set. Should the test dataset be enriched with additional samples to follow the evolution of the training set, a versioning system will be used to ensure the traceability and replicability of the results.

Technical Validation

Data Quality Control. To establish robust data quality control, the project began by training volunteer annotators to adhere to a standardized procedure, detailed on the project's GitLab page (https://gitlab.com/nbm.challenge/procedure). This foundational step ensured consistency in data collection and submission. The automatic validation checks of the web platform (see Section Methods) then helped prevent format mismatch and annotation irregularities, while the manual review ensured the correct placement of bounding boxes and the accuracy of species labelling.

The second phase of validation involved training a machine learning model on the dataset, a procedure detailed in the next section. This provided insights regarding the overall database quality, but also about more specific issues such as class imbalance, heterogeneity of recording quality, or sensitivity to noisy inputs.

The NBM Detection Model: an Object Detection Baseline. The original annotation procedure of the NBM dataset allows the training of object detection models able to precisely localize bird vocalizations in an audio sample, both in time and frequency. This approach makes the enumeration of calls possible, potentially allowing to single out individuals, thereby opening up a range of downstream applications.

This is noticeably different from the usual multi-label classification approach followed for avian sounds[26–28,34,50], which only predicts the presence or absence of a given species in a window of several seconds of audio. An example of output from our model on the NBM test set can be found in Fig. 2, displaying the time and frequency localization of detected calls.

In this section, we present the two-stage model that we trained on the NBM dataset and which performs object detection on spectrograms. Following the description of the model architecture, we present the results of our quantitative evaluations, including a comparison against BirdNet[26] on a multi-label classification task using sliding windows. Our model's highly competitive results across most species highlight the effectiveness of our framework and underline the value of the NBM dataset in advancing automatic bird sound recognition.

Model Architecture. Object detection neural networks can come either as one-stage[58–62] or two-stage models[46,63–65]. The latter involves a pass through a first network which is tasked with retrieving all potential Regions of Interest (RoIs) in an input image, corresponding to object positions. In a second stage, another network performs classification on the RoIs. One-stage object detectors, on the other hand, directly predict a class for each position, or anchor, in the image, including an extra background class. Both approaches rely on an initial feature extraction through a backbone network, whose depth and expressive power directly impact the quality of the detection.

Owing to preliminary experiments, *we find two-stage models to be significantly easier to train* on our avian sound detection dataset. We hypothesize that this is due to the overrepresentation of the background class within samples, and that this class is not always correlated with an absence of signal: all sorts of background noises, starting with raindrops, will appear as peaks of intensity in the input spectrograms. It is therefore an easier task for the model to tackle avian signal detection and classification as two separate problems.

In light of these findings, we build our detection model on the FPN-FasterRCNN (hereafter FRCNN) architecture[46,47], largely modernized and tailored for the detection of bird calls. The architecture consists of the following blocks, and is represented in Fig. 3.

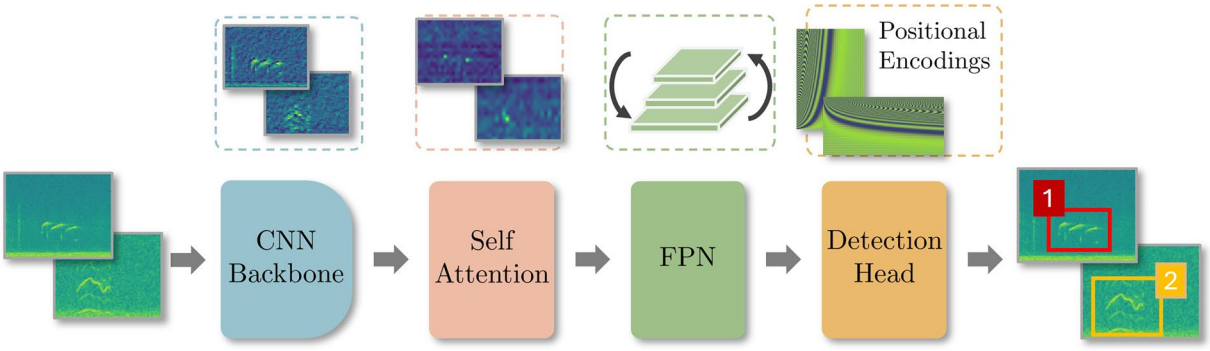


Fig. 3 Overview of the presented NBM bird sound detection modular architecture.

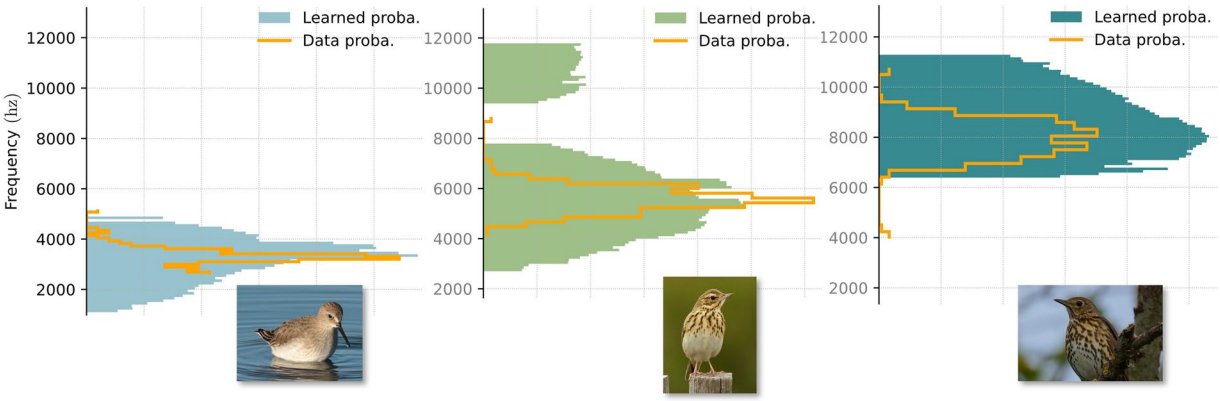


Fig. 4 Learned posterior probabilities for call frequencies vs data distribution on three species: dunlin (left), tree pipit (middle), and song thrush (right). Note the alignment between ground truth and predicted frequencies, learned solely through the use of the positional encoding.

Backbone & FPN. The proposed NBM detection model is built upon a modular architecture where each of the blocks represented in Fig. 3 operates independently from the others, allowing the exploration of diverse architectures within each block. In the following, an EfficientNetV2 backbone[66] is used in place of the VGG16 network of the seminal FRCNN paper. Similarly, one may explore using several variants of the Feature Pyramid Network (FPN,[47]), whose primary goal is to feed lower layers of the network with semantic information through a top-down path[60,67]. We experimentally find that increasing the number of output channels of a single FPN layer yields the best trade-off between accuracy and model complexity.

Self-Attention Module. Contrary to natural images, there is a strong correlation between bird vocalizations that can be found in a given time frame. Equipping the model with an attention mechanism allows each local prediction to leverage the contextual information in which the call was emitted, leading to overall better predictions. This is done by performing self-attention[68,69] separately at each intermediate output from the backbone, prior to the FPN (second block of Fig. 3).

Detection Head. The detection head is composed of two networks: a Region Proposal Network (RPN) and a Regional CNN (RCNN), which processes and classifies the outputs from the RPN (we refer the reader to the original paper for more details). Contrary to other vision tasks, object detection on spectrograms *is not invariant on the size of the object and its position on the frequency axis*: similar patterns emitted at two different fundamental frequencies are likely to belong to two distinct species. To account for this effect, sinusoidal positional encodings are added to the extracted RoIs (fourth block in Fig. 3). We use an absolute positional encoding for the frequency axis and a relative encoding for the time axis.

The effectiveness of this strategy is illustrated in Fig. 4. Here we investigate how the output probability attributed to an input call varies when the frequency encoding changes, which mimics the extraction of the same RoI over different locations on the frequency axis. By doing so, one obtains the conditional classification probabilities $p(c|f)$ for all values of the frequency f, where c is known and is the class of the input call. Bayes' rule then provides the posterior probability $p(f|c)$ on the frequency, learned by the model only through the use of the positional encoding:

Model	GrGr	ErRu	FrCo	MeNi	HaOs	TuPh	AnTr	BuOe	ChHi	MoFl	TuIl	RaAq	PaMa	VaVa	StAl
NBM	0.76	0.33	0.54	0.90	0.89	0.93	0.84	0.76	1.00	0.97	0.95	0.72	0.57	0.69	0.61
BirdNet	0.80	0.42	0.58	0.99	0.89	0.52	0.84	0.86	0.98	0.87	0.84	0.44	0.60	0.79	0.82
GaCh	AnCr	AcHy	FiHy	EmHo	PlSq	CoCo	AtNo	FuAt	AlAr	ClGl	CyCa	EmCi	TrTo	NuAr	MuSt
0.37	0.99	0.88	0.57	0.70	0.83	1.00	0.38	0.64	0.99	0.87	0.65	0.31	1.00	0.58	0.58
0.70	0.95	0.95	0.24	0.58	0.95	1.00	0.75	0.69	0.80	0.70	0.96	0.80	0.86	0.90	0.61
TyAl	CaAl	TrOc	ArCi	TuMe	ChDu	TaRu	PiPi	PlAp	NuPh	TuTo	AnPr	IxMi	NyNy		**mAP**
0.97	0.69	0.78	0.90	0.3	0.79	0.37	0.67	0.26	1.00	0.69	0.60	0.73	0.59		**0.71**
1.00	0.99	0.99	0.98	0.51	0.80	0.66	1.00	0.71	0.88	0.73	0.69	0.61	1.00		**0.78**

Table 1. Multi-label classification average precision on the considered scope of 45 species for the NBM and BirdNet models.

$$p(f|c) = \frac{p(c|f)p(f)}{p(c)},$$

where $p(c)$ is obtained by marginalizing over f, with all marginals $p(f)$ taking the same value. We report in Fig. 4 posterior probabilities averaged over all test samples for three distinct bird species. The first observation is that these probabilities are not uniform, which shows that the positional encoding indeed biases the model's outputs. More importantly, the posterior probabilities peak at the same frequency values as the empirical distributions derived from the training set, indicating that the model successfully learns to leverage the information contained in the encoding to improve its predictions.

Model Evaluation. We experimentally evaluate the performances of our detection model trained on the NBM dataset. Beyond the validation of the architectural choices presented above, these experiments aim to show the relevance of the proposed dataset for the task of precisely localizing migratory bird calls in acoustic recordings.

Evaluation Scope. As described in the Methods section, collecting the NBM dataset is an ongoing project, where the annotation effort is partly driven by the frequently re-evaluated needs for species suffering from a lack of data samples or diversity. For a fair evaluation, it is therefore necessary to restrict the scope of the study to species that present both a sufficient number of data samples and that appear in a sufficient number of input files. This avoids the biases caused by the over-representation of a small number of individuals in a few long training files. We set the minimal thresholds to 100 training samples and 20 source files, to which we add three migratory species that closely approach the line (*Ixobrychus minutus* & *Anthus pratensis* with 96 samples, and *Nycticorax nycticorax* with 18 files and 85 samples) for a total of 45 species, which can be found in Table 2.

Technical Details. All recordings were resampled to 44100 Hz, and subsequently processed through Short-Time Fourier Transform using 1024 points and a hop size of 132. The detection model's input spectrogram resolution was then set to 375×1024, corresponding to a frequency range between 500 Hz and 13000 Hz and a duration of 3.07 s. Such a large input size is susceptible to provide a richer acoustic context, which the model is able to exploit through its self-attention layer. As we consider objects of smaller size than for the typical detection task, we start extracting activation maps from the second intermediate layer $\mathcal{P}_2$ of the backbone (corresponding to a reduction by a factor 2 of the input spatial dimensions), up until $\mathcal{P}_6$ (factor 2^5, respectively). Finally, we experimentally found that, compared to the original FRCNN configuration, additional anchor aspect ratios (akin to[59]) and larger RPN/RCNN batch sizes provided further improvements.

Performances on the Detection Task. We calculate the mean average precision (mAP) over the 45 target species and the 271 hand-annotated XC recordings that compose the NBM test set. The mAP calculation first involves averaging separately for each species the model precision over all values of recall in the [0, 1] interval, discretized beforehand into a fixed number of bins. A minimal intersection-over-union (IoU) threshold between a predicted and a ground truth bounding box is required to consider a prediction positive. Finally, the obtained average precisions (AP) are averaged over species to give the final mAP value.

We found a **mAP value @IoU$_{0.5}$ of 0.67**, which will constitute the baseline to track future improvements of the model.

Comparison with BirdNet. In this section, the evaluation is carried out on a multi-label classification formulation of the problem, which allows to compare the performances of the NBM model with the state-of-the-art BirdNet model. BirdNet builds on an EfficientNet[70] backbone, and is trained on a dataset that includes the whole XC database to predict the presence of bird species within 3-second time frames. As our test set originates from XC itself, a degree of overlapping with BirdNet's training base is possible, and therefore the results presented here should be treated with caution. However they still provide valuable insights. The two models are not directly comparable as their scope differs, the NBM model tackling bird sound recognition through the object detection

GrGr	Grus grus	ErRu	Erithacus rubecula	FrCo	Fringilla coelebs	MeNi	Melanitta nigra	HaOs	Haematopus ostralegus
TuPh	Turdus philomelos	AnTr	Anthus trivialis	BuOe	Burhinus oedicnemus	ChHi	Charadrius hiaticula	MoFl	Motacilla flava
TuIl	Turdus iliacus	RaAq	Rallus aquaticus	PaMa	Parus major	VaVa	Vanellus vanellus	StAl	Strix aluco
GaCh	Gallinula chloropus	AnCr	Anas crecca	AcHy	Actitis hypoleucos	FiHy	Ficedula hypoleuca	EmHo	Emberiza hortulana
PlSq	Pluvialis squatarola	CoCo	Coturnix coturnix	AtNo	Athene noctua	FuAt	Fulica atra	AlAr	Alauda arvensis
ClGl	Clamator glandarius	CyCa	Cyanistes caeruleus	EmCi	Emberiza citrinella	TrTo	Tringa totanus	NuAr	Numenius arquata
MuSt	Muscicapa striata	TyAl	Tyto alba	CaAl	Calidris alpina	TrOc	Tringa ochropus	ArCi	Ardea cinerea
TuMe	Turdus merula	ChDu	Charadrius dubius	TaRu	Tachybaptus ruficollis	PiPi	Pica pica	PlAp	Pluvialis apricaria
NuPh	Numenius phaeopus	TuTo	Turdus torquatus	AnPr	Anthus pratensis	IxMi	Ixobrychus minutus	NyNy	Nycticorax nycticorax

Table 2. Table of correspondence between species codes and Latin names.

paradigm. It is possible however to cast its outputs in the multi-label classification setting of BirdNet by aggregating all detected sound events in a given window and retaining the maximum detection confidence for each species. The resulting AP scores per species and overall mAP are reported in Table 1. Despite the orders of magnitude of difference in the scale of their respective training sets, the NBM model results are highly competitive with those of BirdNet with 0.07 points of difference in mAP. These results highlight the potential of a precisely annotated, medium-scale dataset (the average number of individual sound events for these 45 species is 306) compared to a much larger-scale, weakly annotated one. This suggests that limited, well-thought-out annotation efforts can have an important and positive impact on the model's performance.

Usage Notes

The results from Table 1 display a high AP variance among species, which mainly illustrates the sensitivity of our model to out-of-distribution test samples, e.g. in the case of *Athene noctua*, *Emberiza citrinella*, *Turdus merula* or *Pluvialis apricaria*. For these species we observe a detrimental covariate shift, arising as a consequence of a mismatch between training and test distributions. Here it may be an indication of a lack of diversity in the recording conditions. Such observations are crucial to better drive the future annotation effort towards a higher intra-class diversity of training samples. A second avenue of improvement is the species coverage which, when insufficient, can be a source of false positive detections as the model assigns even unknown vocalizations to the most likely class seen during training. To mitigate this, we aim to increase the current scope of species to meet the threshold described in the Model Evaluation section for all nocturnal migratory birds of Europe, along with common species that are susceptible to vocalizing during nighttime. Finally, note that unlike the training set, our test set is exclusively composed of focal recordings, which, although presenting various levels of background noise, may not be representative of typical PAM recording conditions dominated by noise

Data availability

All data mentioned in this work, and which support the results presented here are freely accessible at the following address: https://zenodo.org/records/17573913.

Code availability

The code for downloading XC samples and for training or fine-tuning the NBM object detection model is accessible on the Github page of the project, https://github.com/LouisBearing/BirdSoundClassif, and on the following Zenodo repository: https://zenodo.org/records/15662728.

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Author contributions

A.P. and J.L. managed the data collection project and were in charge of the animation of the annotator community. A.P. supervised the development of the web platform on which recordings were uploaded. L.A. reviewed the quality of the annotations and was in charge of the development and training of the object detection model. All three authors were involved in the writing of the present article.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to L.A. or A.P.

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